

Design notebook CS1450 spring-25

Assignment 1

Assignment 1 (cont.)

Step 1: Problem statement (2402) 6 ppt

In main execute 3 tasks. t_1, t_2, t_3

1. Create an Array & fill it with the supplied values
2. Write values into the array to a file in sorted order
3. Read the values in the file & fill even & odd arrays with no duplicates

Step 2: Understandings

What do I know: two

1.074.0 - original values for initial array

to convert logarithmic to

- I know how to create arrays

Step 3: Parse java object into min

→ Sports [file name is given]

P(1), 8, 10, 11, P, P, 12, 8, 21, 12, 01, 31, 12

- I know how to create and read FXML files in java

read from files in java

- I know there is a sort method

- I know there is a

in java for arrays

for array

9/5/77 07. I know how to use indexes

to reference locations in an array

Assignment 1 (cont.)

Step 2 (cont.)

What I don't know / questions

1. It's been a while since I've coded anything and need to get back in the mindset

2. Don't know if there is a java method to remove duplicates or if I have to step through values

3. Don't want to step through every value in an array to check for duplicates everytime I add a new value

step 3: pseudo code for main

Create and populate `int[] array1 =`

`{ 1, 18, 10, 2, 16, 8, 15, 9, 9, 17, 14, 18, 1, 19 }`

Print 1st values

`Arrays.sort` the values in `array1`

Create a file `fileName = new File("Assignment1.txt")`

Open file `PrintWriter outputFile = new PrintWriter`

write all values `array1` to file

Assignment 1 (cont.)

Step 3 (cont.)

Create even counter i (and odd counter j)

Loop through values

$\text{count} \leftarrow \text{count} + 1$ (even $\Rightarrow \% 2 == 0$)

$\text{count} \leftarrow \text{count} + 1$ (odd $\Rightarrow \% 2 == 1$)

$\text{Total} = \text{even} + \text{odd}$

$\text{Print} \text{Total}$

Print $\#$ even values

Print $\#$ odd values

Print $\#$ odd values

Print $\#$ file is in directory (FileName.getAbsoluteFile())

Print $\#$ file is in directory

// Open file for reading

Scanner readFile = new Scanner(FileName)

Scanner readFile = new Scanner(FileName)

Create even array $\text{Int}[]$ length i

Create odd array $\text{Int}[]$ length j

Print $\#$ even values

Loop Values for $n < (i+j)$

// Check even

if $\text{value} \% 2 == 0 \Rightarrow \text{even}$

else $\text{value} \% 2 == 1 \Rightarrow \text{odd}$

if $\text{value} \% 2 == 0 \Rightarrow \text{even}$

else $\text{value} \% 2 == 1 \Rightarrow \text{odd}$

if value is same as previous

index, it's duplicate, don't write

Index

Print even array now dup with 0's

Print odd array now dup with 0's

Print

Assignment 1 (cont.)

Step 3 (cont.)

Create/Initialize even/odd counters
Loop through values
If even = non-dup $i++$ same value
If odd non-dup $j++$ same value
Create final arrays
Loop & save values
Print final arrays non-dup
No zeros at end

Step 4: Lessons Learned

1. By counting the non-dups earlier it made creating & populating right-sized arrays a lot easier
2. Couldn't easily find a non-dup method but really didn't need one since the values were sorted before removing dup. Just compared to $\text{Index} - 1$
3. Didn't need all exception handlers I imported